

3.7 Rigid Bodies

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs F13–F16

What this topic covers. Moments of forces, the two conditions for a rigid body to be in equilibrium, the weight acting through the centre of mass, and problems involving rods and laminas resting on supports.

The scope of this section. MEI restricts it to bodies modelled as *possibly non-uniform rods* and **rectangular laminas**, with **coplanar** forces acting **perpendicular** to the rod or to an edge of the lamina. A **vector treatment of moments** and finding the **centre of mass of a composite body** are both **excluded**.

1. Why a rigid body is different

A particle has no size, so all forces act at one point and can only produce translation. A **rigid body** has extent, so where a force is applied matters: a system of forces can have a **turning effect** (spec ref F15).

2. Moments

The **moment** of a force about a point is

$$\text{moment} = \text{force} \times \text{perpendicular distance from the point}$$

measured in **newton metres (N m)**.

A moment has a sense of rotation — clockwise or anticlockwise — which is why the direction must be stated or a sign convention adopted.

Worked example. A force of 12 N acts perpendicular to a rod, 0.5 m from the pivot.

$$\text{moment} = 12 \times 0.5 = 6 \text{ N m}$$

A 15 N force at 2 m gives a moment of 30 N m; a 40 N force at 3 m gives 120 N m.

A force acting **through** the pivot has zero moment about it, because its perpendicular distance is zero. This is what makes the choice of pivot so useful — see §5.

Although N m and joules share base units, a moment is always quoted in **N m**, never in joules.

3. Equilibrium of a rigid body

A rigid body is in equilibrium when **both** conditions hold (F14):

1. The resultant force is zero.
2. The sum of the moments about any one point is zero.

Two conditions means two kinds of equation — resolving, and taking moments. For a body under parallel vertical forces this gives:

$$\text{total upward force} = \text{total downward force}$$

$$\text{total clockwise moment} = \text{total anticlockwise moment}$$

Both conditions are needed. A body can have zero resultant force yet still rotate — two equal and opposite forces applied at different points along a rod produce no net force but a definite turning effect.

Because the sum of the moments is zero about **any** point once the body is in equilibrium, you may choose whichever point is most convenient.

4. Weight and the centre of mass

For the purpose of calculating its moment, the weight of a body may be taken to act through a single point — its **centre of mass** (F16).

- For a **uniform** rod, the centre of mass is at the **midpoint**. A uniform rod of length $2a$ has its weight acting a from either end.
- For a **uniform rectangular lamina**, it is at the intersection of the diagonals.
- For a **non-uniform** rod, it is *not* at the midpoint. It will either be given, or found from symmetry, or found by taking moments.

5. Choosing the pivot

This is the single most useful technique in the topic. Take moments about a point through which an **unknown force acts** — that force then has zero moment and disappears from the equation, leaving one unknown instead of two.

Worked example. A uniform plank of weight 200 N and length 4 m rests on supports at each end, with a 100 N load at its midpoint. Find each reaction.

By symmetry the two reactions are equal. Resolving vertically:

$$R_1 + R_2 = 200 + 100 = 300 \implies R_1 = R_2 = 150 \text{ N}$$

Worked example. A light rod is pivoted at its centre. A 30 N weight hangs 2 m from the pivot. At what distance must a 40 N weight hang on the other side to balance it?

$$30 \times 2 = 40 \times d \implies d = 1.5 \text{ m}$$

Worked example. A uniform rod of weight 60 N and length 4 m is hinged at A and held horizontal by a vertical string at B . Find the tension and the vertical force at the hinge.

Taking moments about A eliminates the hinge force. The weight acts at the midpoint, 2 m from A :

$$T \times 4 = 60 \times 2 \implies T = 30 \text{ N}$$

Resolving vertically: the hinge force is $60 - 30 = 30 \text{ N}$ upwards.

6. Non-uniform rods

When the rod is non-uniform, the position of the centre of mass becomes the unknown, and moments are used to locate it.

Worked example. A non-uniform rod AB of length 4 m and weight 50 N rests horizontally on supports at A and B . The reaction at A is 20 N. Find the distance of the centre of mass from A .
Resolving vertically: $R_A + R_B = 50$, so $R_B = 30 \text{ N}$.

Taking moments about A , with the centre of mass at distance x :

$$50x = 30 \times 4 = 120 \implies x = 2.4 \text{ m}$$

Worked example. A non-uniform rod of weight 80 N and length 5 m rests on supports at its ends, with reactions 30 N and 50 N. Find the centre of mass from the 30 N end.

$$80x = 50 \times 5 = 250 \implies x = 3.125 \text{ m}$$

Notice the pattern: the centre of mass sits **closer to the larger reaction**. Use that as a sanity check — an answer placing it nearer the smaller reaction is wrong.

7. Tipping

A body resting on two supports is on the point of **tipping** about one of them when the reaction at the *other* support falls to **zero**.

That condition converts a tipping question into an ordinary moments problem: set the far reaction to zero and take moments about the remaining support.

Worked example. A uniform plank of length 6 m and weight 300 N rests on supports 1 m from each end, so the supports are at 1 m and 5 m. A child of weight 200 N stands at one end. Find the reaction at the nearer support.

Taking moments about the far support at 5 m, with the child at 0 m and the weight at the centre, 3 m:

$$R \times 4 = 300 \times 2 + 200 \times 5 = 600 + 1000 = 1600 \implies R = 400 \text{ N}$$

8. Setting out a solution

1. Draw the rod or lamina with **every** force marked, including the weight at the centre of mass.
2. Resolve vertically to relate the reactions to the total load.
3. Take moments about a well-chosen point to eliminate an unknown.
4. Solve, then sanity-check the magnitudes and positions.

For a body in equilibrium under vertical forces there are two independent equations available, so at most two unknowns can be found.

9. Common pitfalls

- Using the distance along the force rather than the **perpendicular** distance.
- Forgetting the weight, or placing it at the midpoint of a *non-uniform* rod.
- Using only the moments condition and forgetting to resolve, or the reverse.
- Choosing a pivot that leaves two unknowns in the equation.
- Mixing clockwise and anticlockwise moments without a consistent sign convention.
- Quoting a moment in joules.
- Setting the wrong reaction to zero in a tipping problem — it is the one at the support the body lifts away from.

- Attempting a vector treatment of moments, or a composite centre of mass, both of which are off-specification.